NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

End-Semester Exam December 2025

Subject: DATABASE MANAGEMENT SYSTEM LAB (CS34109)

Full Marks: 60 **Duration: 3 Hours**

Answer all questions. Make assumption of missing data if any.

Please write precise and to the point answers, irrelevant and lengthy answers may attract penalty.

S. No.		Question	Marks	CO	BL
1	a)	Consider the Universal relation R = (A, B, C, D, E, F, G, H, I, J) and set of functional dependencies F = { {A, B} → C , {A} → {D, E}, {B} → {F}, {F} → {G, H}, {D} → {I, J} What is the key of R? Decompose R into 2NF and then 3NF relation.	5	CO6	Apply
	b)	Consider the following decompositions for the above relational schema R. Determine whether each decomposition has (i) Dependency preserving property, (ii) Lossless join property Functional dependencies given (Q.1.A). i) D_1 = { R_1, R_2, R_3, R_4, R_5 } R_1 = { A, B, C }, R_2 = { A, D, E }, R_3 = { B, F }, R_4 = { F, G, H }, R_5 = { D, I, J } ii) D_2 = { R_1, R_2, R_3 } R_1 = { A, B, C, D, E }, R_2 = { B, F, G, H }, R_3 = { D, I, J }	4	CO6	Apply
	c)	Consider a relation R_1 (A, B, C, D, E) with the following dependencies AB → C, CD → E, DE → B Is AB a candidate key of this relation? If not, is ABD? Explain your answer. Explain the reflexivity and the augmentation rule.	4+2	CO5, CO6	Apply
2	a)	"All BCNF is in 3NF but the reverse is not true." Explain it with proper example.	4	CO6	Understand
	b)	Consider three transactions T_1, T_2 and T_3 and two schedules S_1 and S_2 given below Draw the precedence graph for schedules S_1 and S_2 and test whether they are conflict serializable or not. T_1: $r_1(P)$, $r_1(R)$, $w_1(P)$, T_2: $r_2(R)$, $r_2(Q)$, $w_2(R)$, $w_2(Q)$, T_3: $r_3(P)$, $r_3(Q)$, $w_3(Q)$; **Schedule S_1**: $r_3(P)$; $r_2(R)$; $r_1(R)$; $r_3(P)$; $r_3(Q)$; $w_1(P)$; $w_3(Q)$; $r_2(Q)$; $w_2(R)$; $w_2(Q)$ **Schedule S_2**: $r_1(P)$; $r_2(R)$; $r_3(P)$; $r_1(R)$; $r_2(Q)$; $r_3(Q)$; $w_1(P)$; $w_2(R)$; $w_3(Q)$; $w_2(Q)$	6	CO8	Apply
	c)	Explain with example disadvantage of concurrent transactions. Explain the view serializability in detail.	5	CO8	Remember
3	a)	Explain Query Optimization with the help of Block Diagram.	3	CO7	Understand
	b)	Find the SQL query, relational algebra and query tree as well as query optimized tree of the following query EMP (Ename, Street, City) Works (Ename, Company_name, Salary) Company (Company_name, City) i. Find the name, street, city of all employees who work for "FBC" company and earn more than ₹10,000 ii. Find the name and salary of all employees who live in the same city as the company for which they work.	6+6	CO7	Apply
4	a)	Consider the following tables Emp (name, eno, dob, address, sex, sal, supereno, dno) Dept (dno, dname, mgreno, mgrstartdate) Dept_loc(dno, dlocation) Project(Pname, Pno, Ploc, dno) Workson(eno, Pno, hrs) Dependent(eno, dependentname, sex, birthdate, relation) Specify the relational algebra and SQL of the following query. i. Retrieve the name of employees in department 5 who work more than 10 hrs per week on the "Development" project. ii. Find the name of all employees who are directly supervised by "Ram". iii. Retrieve the average salary of female employees. iv. For each project, list the project name and total hours per week spent on the project. v. Retrieve the name of employees who have no dependent.	15	CO3	Apply